

**A Sprint** fixed period or duration in which a team works to complete a set of tasks

**An Epic** is a **big task or project** that is too large to complete in one sprint. It is broken down into **smaller tasks (stories)** that can be completed over multiple sprints.

**A Story** is a small task. It is part of an **Epic**.

**A Story Point** is a number that represents how much effort a story takes to complete. (usually in form of Fibonacci series)

- Very Easy task
- Normal task
- Moderate task
- Difficult task

**Storypoint -5 (1,2,3,5)**

**Sprint 1**

**Data Collection and Analysis (Epic 2)**

**Download and collect agricultural dataset (USN1) 2**

**Import required Python libraries (USN2) 1**

**Read and explore dataset structure and features (USN3) 2**

**Perform univariate analysis on N, P, K, temperature, humidity, pH, rainfall (USN4) 3**

**Conduct bivariate analysis on agricultural parameters vs crop suitability (USN5) 3**

**Perform multivariate analysis to discover patterns among variables (USN6) 3**

**Define Problem and Understanding (Epic 1)**

**Identify and define the agricultural problem for OptiCrop (USN7) 2**

**Gather and analyze business requirements and project objectives (USN8) 1**

**Conduct literature survey on crop recommendation techniques (USN9) 3**

**Analyze social and business impact of AI-driven crop recommendation (USN10) 3**

**Total Story Point in Sprint 1= 2+1+2+3+3+3+2+1+3+3=23**

## **Sprint 2**

### **Data Pre-Processing (Epic 3)**

**Check for null values and handle missing data (USN11) 2**

**Detect and treat outliers affecting model performance (USN12) 2**

**Extract seasonal crop information and prepare relevant features (USN13) 3**

**Split dataset into training and testing sets (USN14) 1**

### **Model Building (Epic 4)**

**Apply K-Means Clustering to identify patterns in agricultural conditions (USN15) 4**

**Train Logistic Regression model for crop prediction (USN16) 5**

**Evaluate model performance and compare results (USN17) 3**

**Save best-performing model and generate crop predictions (USN18) 3**

### **Application Building (Epic 5)**

**Design and develop HTML pages for interactive user interface (USN19) 5**

**Build Python backend and integrate with trained ML model (USN20) 5**

**Run, test, and validate the OptiCrop application (USN21) 3**

**Total story Point in Sprint 2= 2+2+3+1+4+5+3+3+5+5+3=36**

## **Total Story Points**

**Sprint 1 = 23**

**Sprint 2 = 36**

**Velocity= Total Story Points Completed/ Number of Sprints**

**Total story Points= 23+36=59**

**No of Sprints= 2**

**Velocity =59/2**

**29.5(Story Points per Sprint)**

**Your team's velocity is 29.5 Story Points per Sprint.**